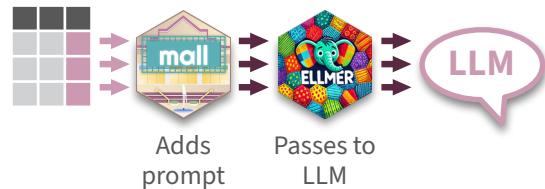


Natural Language Processing using LLMs in R :: CHEAT SHEET



Intro

Use LLM's to perform NLP row-wise over a data frame. **mall** comes with pre-defined prompts that perform specific NLP operations, and then places the results in a new column. Use **OpenAI**, **Ollama**, **Anthropic** and many others thanks to its integration with **eLLMER**:



mall's data frame functions are designed with 'tidy' principals in mind, so they work with **tidyverse** packages.

mall also includes functions that work with string vectors.

Getting Started

1. Load the libraries

```
library(mall)
library(eLLMER)
```

2. Create a vendor specific chat connection

```
chat <- chat_openai()
```

3. Pass the chat object to **mall**

```
llm_use(chat)
```

For dataframes:

```
data("reviews") # Sample product reviews
reviews |>
  llm_sentiment(review)
```

For vectors:

```
llm_vec_sentiment(c("I am happy", "I am sad"))
```

CONNECT AUTOMATICALLY

As a convenience, **mall** is able to automatically establish a connection with the LLM. To do this you can use the **.mall_chat** option:

```
options(.mall_chat=eLLMER::chat_openai(model="gpt-4o"))
```

Add this line to your **.Rprofile** file in order for that code to run every time you start R. You can call **usethis::edit_r_profile()** to edit.

NLP Functions

☆☆☆ SENTIMENT ANALYSIS

```
llm_sentiment(.data, col, options = c("positive",
  "negative", "neutral"), pred_name = ".sentiment",
  additional_prompt = "")
llm_sentiment(reviews, review)
```

```
llm_vec_sentiment(x, options = c("positive",
  "negative", "neutral"), additional_prompt = "",
  preview = FALSE)
llm_vec_sentiment(c("I am happy", "I am sad"))
```

Special arguments:

options - Customize the sentiments to check for: ``options = c("positive", "negative")``. Use 'tilde' to mask the results, for example ``c("positive" ~ 1, "negative" ~ 0)`` returns 1 for positive and 0 for negative.

🔍 EXTRACT

Extract specific entity, or entities, from the provided text

```
llm_extract(.data, col, labels, expand_cols =
  FALSE, additional_prompt = "", pred_name =
  ".extract")
llm_extract(reviews, review, labels = "product")
```

```
llm_vec_extract(x, labels = c(),
  additional_prompt = "", preview = FALSE)
llm_vec_extract("bob smith, 123 3rd street",
  c("name", "address"))
```

Special arguments:

labels - A vector to specify the entities to identify
expand_cols - If multiple labels, this indicates if the labels will show up in their own column (data frames only)

🗨️ SUMMARIZE

Summarize text into a specified number of words

```
llm_summarize(.data, col, max_words = 10,
  pred_name = ".summary", additional_prompt =
  "")
llm_summarize(reviews, review, max_words = 5)
```

```
llm_vec_summarize(x, max_words = 10,
  additional_prompt = "", preview = FALSE)
llm_vec_summarize("This has been the best TV I've
  ever used. Great screen, and sound.", max_words = 5)
```

✓✗ VERIFY

Check if a statement is true or not based on the provided text

```
llm_verify(.data, col, what, yes_no = factor(c(1,
  0)), pred_name = ".verify", additional_prompt =
  "")
llm_verify(reviews, review, "is the customer
  happy")
```

```
llm_vec_verify(x, what, yes_no = factor(c(1, 0)),
  additional_prompt = "", preview = FALSE)
llm_vec_verify(c("I am happy", "I am sad"), "is the
  person happy")
```

Special arguments:

yes_no - Customize what it returns for true/false with a vector ``yes_no = c("y", "n")``.

🔍 CLASSIFY

Classify the provided text as one of the options provided via the 'labels'

```
llm_classify(.data, col, labels, pred_name =
  ".classify", additional_prompt = "")
llm_classify(reviews, review, c("appliance",
  "computer"))
```

```
llm_vec_classify(x, labels, additional_prompt =
  "", preview = FALSE)
llm_vec_classify(c("this is important!", "just
  whenever"), c("urgent", "not urgent"))
```

Special arguments:

labels - A character vector with at least 2 labels to classify the text as

🇪🇺➡🇬🇧 TRANSLATE

Translate into target language

```
llm_translate(.data, col, language, pred_name =
  ".translation", additional_prompt = "")
llm_translate(reviews, review, "spanish")
```

```
llm_vec_translate(x, language,
  additional_prompt = "", preview = FALSE)
llm_vec_translate("grass is green", "spanish")
```

Special arguments:

language - Target language. No origin language is passed since the LLM detects it automatically.

📝 CUSTOM

Create your own prompt

```
llm_custom(.data, col, prompt = "", pred_name =
  ".pred", valid_resps = "")
my_prompt <- "Answer a question. Return only the
  answer, no explanation. Acceptable answers are
  'yes', 'no'. Answer this about the following text, is
  this a happy customer?:"
llm_custom(reviews, review, my_prompt)
```

```
llm_vec_custom(x, prompt = "", valid_resps =
  NULL)
```

Special arguments:

valid_resps - A vector to specify the set of answers expected back. 'mall' will change those not in the set to NA.

SHARED ARGUMENTS

additional_prompt - Appends instructions to the LLM.
`llm_classify(reviews, review, c("appliance", "computers"), additional_prompt = "Consider TVs as appliances")`

pred_name - Name of the new column. Defaults are set based on the NLP operation. (Data frames only)
`llm_vec_translate("grass is green", "spanish", pred_name = "in_spanish")`

preview - Returns what it would be sent to the LLM instead (Vectors only)

Other Features

OLLAMA DIRECT

If Ollama is the only LLM provider you are using, then a simplified way to connect is available which does not require an eLLMER Chat object. Simply pass "ollama" as the 'backend', and specify the model:

```
llm_use("ollama", model= "llama3.2")
```

CACHING RESULTS

By default, **mall** saves the LLM results in a temp folder. To specify a folder call:

```
llm_use(chat, .cache = "<my folder>")
```

To turn off use:

```
llm_use(chat, .cache = "")
```

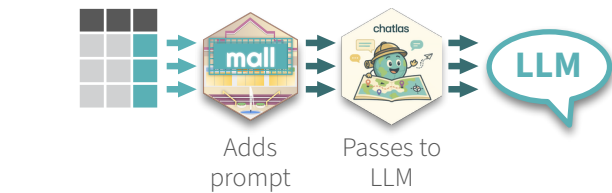


Natural Language Processing using LLMs in Python :: CHEAT SHEET



Intro

Use LLM's to perform NLP row-wise over a data frame. **mall** comes with pre-defined prompts that perform specific NLP operations, and then places the results in a new column. Use **OpenAI**, **Ollama**, **Anthropic** and many others thanks to its integration with **chatlas**:



mall works as an extension for Polars data frames. It also works with string vectors.

Getting Started

Start by creating a new LLM connection

```
from chatlas import ChatOpenAI
chat = ChatOpenAI()
```

DATA FRAMES

1. Load the library

```
import mall
```

2. Read or load your data frame

```
reviews = mall.MallData.reviews # Sample product reviews
```

3. Pass the chat object to mall

```
reviews.llm.use(chat)
```

4. Access NLP functions via .llm

```
reviews.llm.sentiment('review')
```

STRING VECTORS

1. Load the LLMVec class

```
from mall import LLMVec
```

2. Create a new LLMVec object

```
llm = LLMVec(chat)
```

3. Pass a vector to a function in the new object

```
llm.sentiment(['I am happy', 'I am sad'])
```

NLP Functions

★★★ SENTIMENT ANALYSIS

```
<DataFrame>.llm.sentiment(col, options =
['positive', 'negative', 'neutral'], additional='',
pred_name='sentiment')
reviews.llm.sentiment("review")
```

```
<LLMVec object>.sentiment(x,
options=['positive', 'negative', 'neutral'],
additional='')
llm.sentiment(['I am happy', 'I am sad'])
```

Special arguments:

options - Customize the sentiments to check for: `options = ["positive", "negative"]`. Use a DICT object to mask the results, for example `{ "positive" : 1, "negative" : 0 }` returns 1 for positive and 0 for negative.

🔍 EXTRACT

Extract specific entity, or entities, from the provided text

```
<DataFrame>.llm.extract(col, labels="",
expand_cols=False, additional="", pred_name='extract')
reviews.llm.extract("review", labels="product")
```

```
<LLMVec object>.extract(x, labels="", additional='')
llm.extract(['bob smith, 123 3rd street'],
labels=['name', 'address'])
```

Special arguments:

labels - A vector to specify the entities to identify
expand_cols - If multiple labels, this indicates if the labels will show up in their own column (data frames only)

📄 SUMMARIZE

Summarize text into a specified number of words

```
<DataFrame>.llm.summarize(x, max_words=10,
additional='')
reviews.llm.summarize("review", 5)
```

```
<LLMVec object>.summarize(x, max_words=10,
additional='')
llm.summarize(['This has been the best TV I've ever
used. Great screen, and sound.'], max_words=5)
```

✅❌ VERIFY

Check if a statement is true or not based on the provided text

```
<DataFrame>.llm.verify(col, what='', yes_no=[1,
0], additional='', pred_name='verify')
reviews.llm.verify("review", "is the customer happy")
```

```
<LLMVec object>.verify(x, what='', yes_no=[1, 0],
additional='')
llm.verify(['I am happy', 'I am sad'], what='Is the
person happy?')
```

Special arguments:

yes_no - Customize what it returns for true/false with a vector `yes\_no = ["y", "n"]`.

🔗 CLASSIFY

Classify the provided text as one of the options provided via the `labels`

```
<DataFrame>.llm.classify(col, labels="",
additional='', pred_name='classify')
reviews.llm.classify("review", ["appliance",
"computer"])
```

```
<LLMVec object>.classify(x, labels="",
additional='')
llm.classify(['this is important!', 'there is no rush'],
['urgent', 'not urgent'])
```

Special arguments:

labels - A character vector with at least 2 labels to classify the text as

🇪🇺➡🇮🇹 TRANSLATE

Translate into target language

```
<DataFrame>.llm.translate(col, language="",
additional='', pred_name='translation')
reviews.llm.translate("review", "spanish")
```

```
<LLMVec object>.translate(x, language="",
additional='')
llm.translate(['the grass is green'], language='spanish')
```

Special arguments:

language - Target language. No origin language is passed since the LLM detects it automatically.

📝 CUSTOM

Create your own prompt

```
<DataFrame>.llm.custom(col, prompt='',
valid_resps='', pred_name='custom')
my_prompt='Answer a question. Return only the
answer, no explanation. Acceptable answers are
'yes', 'no'. Answer this about the following text, is
this a happy customer?:'
reviews.llm.custom("review", prompt=my_prompt)
<LLMVec object>.custom(x, prompt='',
valid_resps='')
```

Special arguments:

valid_resps - A vector to specify the set of answers expected back.

SHARED ARGUMENTS

additional - Appends more instructions to the LLM.
`reviews.llm.classify("review", ["appliance", "computer"], additional="Consider TVs as appliances")`

pred_name - Name of the new column. Defaults are set based on the NLP operation. (Data frames only)

Other Features

OLLAMA DIRECT

If Ollama is the only LLM provider you are using, then a simplified way to connect is available which does not require an eLLmer Chat object. Simply pass "ollama" as the `backend`, and specify the model:

```
Dataframe:
reviews.llm.use('ollama', 'llama3.2')
Vector:
llm = LLMVec('ollama', 'llama3.2')
```

CACHING RESULTS

By default, mall saves the LLM results in a temp folder. To specify a folder call:

```
Dataframe:
reviews.llm.use(chat, _cache='<my folder>')
Vector:
llm = LLMVec(chat, _cache='<my folder>')
To turn off use:
Dataframe:
reviews.llm.use(chat, _cache='')
Vector:
llm = LLMVec(chat, _cache='')
```

